

ASELE, Sweden

WMO: 022540

Lat: 64.165N

Lon: 17.315E

Elev: 308

StdP: 97.68

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-31.5	-28.0	-34.9	0.1	-31.5	-30.9	0.2	-27.9	7.2	0.6	6.2	-0.1	0.0	110	0.656

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.2	25.4	16.5	23.4	15.9	21.5	14.9	18.2	23.1	16.9	21.7	15.7	19.9	2.2	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.3	12.0	19.7	15.0	11.1	18.5	13.8	10.2	17.6	52.5	22.9	48.5	21.6	45.1	19.9	22.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)		
6.2	5.3	4.5		-34.5	27.6	3.0	2.1	-36.7	29.1	-38.5	30.3	-40.2	31.5	-42.4	33.0	
				-34.8	19.8	2.9	1.5	-36.9	20.9	-38.6	21.8	-40.2	22.7	-42.4	23.8	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	1.2	-11.0	-10.2	-5.6	0.7	6.6	11.1	14.2	12.1	7.4	1.1	-4.2	-8.4
	DBStd	10.11	7.85	7.54	5.95	3.63	3.70	3.38	3.04	3.03	3.16	4.19	5.56	7.61
	HDD10.0	3495	652	564	483	278	118	25	2	13	88	276	425	569
	HDD18.3	6247	911	798	742	528	363	217	132	194	327	533	675	828
	CDD10.0	296	0	0	0	0	13	59	133	79	11	1	0	0
	CDD18.3	6	0	0	0	0	0	1	4	1	0	0	0	0
	CDH23.3	176	0	0	0	0	6	42	106	22	0	0	0	0
	CDH26.7	20	0	0	0	0	0	3	17	1	0	0	0	0
Wind	WSAvg	1.8	1.4	1.5	2.0	1.9	2.2	2.3	1.8	1.6	1.8	1.7	1.5	1.4
Precipitation	PrecAvg	608	45	30	28	30	42	64	88	85	54	52	48	42
	PrecMax	824	87	65	58	67	74	119	172	167	143	138	99	80
	PrecMin	419	4	3	14	5	10	14	24	23	13	17	20	5
	PrecStd	104	18	15	11	16	20	27	37	45	30	26	25	16
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.7	5.3	7.7	17.2	24.1	26.9	28.8	25.7	19.2	13.1	8.6	4.7
		MCWB	3.0	3.3	4.0	8.9	13.7	15.9	18.6	17.7	14.0	9.9	7.6	3.4
	2%	DB	2.7	3.9	5.6	12.5	21.1	24.5	26.0	23.5	16.7	10.1	5.6	3.2
		MCWB	1.7	2.1	2.6	6.6	12.4	14.8	17.4	16.7	13.0	8.5	4.9	2.1
	5%	DB	1.0	1.8	4.2	10.0	18.5	21.5	24.1	21.4	14.8	8.8	3.7	2.0
		MCWB	0.2	0.7	1.9	5.1	11.3	13.9	16.8	15.6	11.8	7.5	3.1	1.3
	10%	DB	-0.4	0.0	2.8	7.7	15.5	19.2	22.1	19.0	13.3	7.6	2.5	0.8
		MCWB	-0.8	-0.8	0.9	3.9	9.5	12.6	15.8	14.6	11.0	6.5	1.9	0.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.2	3.4	4.6	9.2	14.9	17.6	20.1	19.1	15.1	10.2	7.7	3.7
		MCDB	4.5	4.9	7.6	16.0	21.2	24.4	25.2	23.9	17.6	11.8	8.6	4.5
	2%	WB	1.8	2.2	3.1	7.3	13.4	15.7	18.7	17.5	13.5	8.9	5.0	2.2
		MCDB	2.7	3.6	5.1	12.0	19.8	21.8	23.4	21.6	15.9	9.8	5.6	2.9
	5%	WB	0.4	0.8	2.1	5.6	11.7	14.5	17.6	16.2	12.4	7.7	3.3	1.3
		MCDB	0.9	1.7	3.9	9.2	17.6	20.0	22.6	19.9	14.4	8.7	3.7	1.9
	10%	WB	-0.9	-0.8	1.1	4.1	9.9	13.2	16.4	15.1	11.4	6.5	2.0	0.3
		MCDB	-0.4	-0.1	2.6	7.3	14.8	18.2	21.0	18.5	13.1	7.5	2.4	0.8
Mean Daily Temperature Range	5% DB	MDBR	9.2	10.3	11.6	10.8	12.1	11.7	12.2	11.3	9.7	7.2	6.2	8.0
		MCDBR	8.5	8.3	10.3	15.5	18.4	16.4	16.2	15.8	12.4	8.7	5.7	7.5
		MCWBR	8.1	7.2	7.8	9.9	10.3	8.1	8.5	9.4	8.7	7.0	5.2	7.0
	5% WB	MCDBR	8.7	8.0	9.6	13.9	17.1	14.8	14.1	13.7	10.9	8.0	5.4	7.4
		MCWBR	8.3	7.1	7.6	9.1	10.0	7.8	8.0	8.5	8.0	6.7	5.0	7.0
Clear-Sky Solar Irradiance	taub	0.196	0.265	0.292	0.328	0.333	0.332	0.345	0.333	0.305	0.288	0.345	0.351	
	taud	1.993	2.206	2.268	2.339	2.475	2.501	2.479	2.502	2.568	2.455	2.758	2.758	
	Ebn at Noon	387	657	799	850	881	887	863	841	800	645	370	207	
	Edh at Noon	26	57	82	97	93	93	93	83	63	46	23	15	
All-Sky Solar Radiation	RadAvg	0.15	0.74	2.15	3.83	4.72	5.31	4.85	3.65	2.13	0.89	0.22	0.06	
	RadStd	0.01	0.11	0.23	0.31	0.55	0.40	0.54	0.36	0.25	0.14	0.04	0.01	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (44 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	-134	N/A	N/A	N/A

Nomenclature: See separate page